

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

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Experiment 8

Effect of PCA on Regression and Classification Models

Objective

To study the impact of **Principal Component Analysis (PCA)** on the performance of machine learning models.

- a) Train models in the original feature space (Without PCA)
- b) Train models in reduced feature space (With PCA)
- c) Compare performance using 5-fold cross-validation

Dataset

Dataset: PCA_Full_Lab_Dataset_1000_Samples.csv

- Total samples: 1000
- Total features: 11 (Math, Physics, Chemistry, Biology, Total_Science, Average_Science, English, Computer_Science, Sports_Score, Attendance_Percentage, Random_Noise_Feature)
- **Regression Target:** Final_Score_Regression (continuous)
- **Classification Target:** Performance_Level_Classification (High / Medium / Low)
- Class distribution: High=202, Low=253, Medium=545
- Missing values: None

Preprocessing Steps

1. No missing values found; no imputation required.
2. Encode classification target with LabelEncoder: Low=0, High=1, Medium=2.
3. Standardise all features using **StandardScaler** (zero mean, unit variance) – *critical before PCA*.
4. Evaluate using 5-fold cross-validation (KFold for regression, StratifiedKFold for classification).

PCA Implementation

1. Apply PCA after standardisation.
2. Retain 95% of explained variance (automatic component selection).
3. Plot Scree Plot showing cumulative variance vs. number of components.

Table 1: PCA Summary

Components Chosen	Explained Variance (%)	Justification
7 out of 11	95.90%	Retains 95% variance threshold; removes 4 redundant/noisy d

Regression Results

Table 2: 5-Fold CV Results – Regression (1000 samples)

Model	Metric	Fold Avg (No-PCA)	Fold Avg (With-PCA)	Observation
Linear Regression	MSE	26.2338	26.2251	Negligible improvement
Linear Regression	R^2	0.7479	0.7481	Nearly identical
Random Forest	MSE	29.0245	28.9563	Slight improvement
Random Forest	R^2	0.7215	0.7222	Minimal PCA effect

Classification Results

Table 3: 5-Fold CV Results – Classification (1000 samples)

Model	Metric	Fold Avg (No-PCA)	Fold Avg (With-PCA)	Observation
Logistic Regression	Accuracy	0.7460	0.7460	No change
Logistic Regression	F1-score	0.7437	0.7438	Identical
SVM (RBF)	Accuracy	0.7390	0.7340	Slight decrease
SVM (RBF)	F1-score	0.7324	0.7264	Slight decrease

Analysis Questions

1. **Did PCA improve Linear Regression?** Negligibly (MSE 26.2338 $\rightarrow$ 26.2251). Since features were already clean and correlated, PCA had little effect on a linear model.

2. **Did PCA significantly affect Random Forest?** No. Tree-based models handle high-dimensional and correlated features natively by splitting on the most informative features, so PCA provides minimal benefit.
3. **Which classification model benefited more from PCA?** Neither model benefited. Logistic Regression was unchanged; SVM slightly decreased ($0.7390 \rightarrow 0.7340$), possibly because PCA's rotation changed the RBF kernel's distance calculations.
4. **Did PCA reduce variance across folds?** Marginally for regression (slight MSE reduction), but not significantly for classification in this dataset.
5. **Why do linear models benefit more from PCA?** Linear models struggle with multicollinearity; PCA decorrelates features and removes noise dimensions. However, with only 11 features and moderate correlation, the gain was minimal here.

Expected Learning Outcome

- PCA reduces multicollinearity and noise dimensions ($11 \rightarrow 7$ components).
- PCA helps linear models more than tree-based models.
- For this dataset, features were already informative; PCA offered negligible benefit.
- Dimensionality reduction is most valuable with many correlated or noisy features.
- Always standardise before PCA.

Final Conclusion

PCA compressed 11 features to 7 components retaining 95.90% variance. For this dataset, PCA had negligible impact on all four models because the original features were already well-conditioned. PCA is most useful when: (1) features are highly correlated, (2) there are many noisy dimensions, or (3) computational cost reduction is critical. It is least necessary for tree-based models (Random Forest, Gradient Boosting) which handle correlated features inherently. The trade-off: PCA reduces dimensionality at the cost of interpretability – components are linear combinations with no direct real-world meaning.